# Technical Reference Manual

## MagicBox v5.8 — hardware, firmware, data flow

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Firmware • Sensors • Data Flow • Games • Exercises

Physics formulas and software implementation

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Firmware version: **2.3.0** (`serial_wifi`, USB + Wi-Fi)
Application version: **v5.4** or later

Last updated: **September 2026**

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# Chapter 1

# Overview

**MagicBox** units are educational devices built around the **ESP32** microcontroller, fitted with four physical sensors. Each box streams data in real time to a desktop application (`PyQt6` + `matplotlib`), where students can visualise, analyse and export it.

The application starts **in English**; Romanian, French or German is chosen in **Settings** → **Language** (the change applies after a restart). The guides and manuals exist in the same four languages. Internal identifiers (sensor keys, module names, the Wi-Fi discovery token) are not translated and are the same in every language.

## 1.1 System architecture

| Layer | Component | Role |
|---|---|---|
| Hardware | ESP32 + sensors | Physical data acquisition |
| Firmware | `serial_wifi` 2.3.0 | Sensor reading, JSON |
| Transport | USB-Serial 115200 baud; Wi-Fi (TCP) | Data transmission |
| Reception | `SerialManager`, `WifiManager` | One thread per link |
| Storage | `DataManager` | Circular buffer + CSV logs on disk |
| Display | PyQt6 + matplotlib | Games, exercises, apps |

## 1.2 Main components

- **4 sensors**: MPU-6500 (accelerometer + gyroscope), BME280 (temperature, humidity, pressure), VEML7700 (light), SCD41 ($CO_2$); the box has **three sockets**, so at most three of them are fitted at any one time (section 2.1)
- **15 guided experiments** in physics, chemistry, biology and geography
- **8 explore tools** (Measurement Notebook, Weather Station, Air Quality, Solar Energy, Seismology, Heat Islands with Several Boxes, The Greenhouse Effect, Spectroscopy)
- **4 games**: Among the Clouds, Earthquake!, Resonance, Gyro Target

The application's **Activities** page shows them in these three groups (**Guided experiments**, **Explore tools**, **Games**); the **Open guide** button on each card opens the teacher guide or the pupil sheet from the PDFs shipped with the application. Each card also names the sensors the activity needs, by what they measure rather than by chip: *Light sensor*, *Temperature, humidity and pressure sensor*, *$CO_2$ sensor*, *Motion sensor* (several are separated by " · ").

## Chapter 2

# Hardware and sensors

### 2.1 Three sockets for four sensors

The box has **three sockets** for sensors, and there are four sensors: light (VEML7700), weather (BME280), $CO_2$ (SCD41) and motion (MPU-6500). So **at most three** are fitted at any one time. The teacher plugs in the sensor the activity needs; the one left out is not a fault.

- At power-on the firmware looks for each sensor once and reports it `OK` or `FAIL`. The absent one shows `FAIL` (for the MPU-6500, for example, `[MPU6500] WHO_AM_I=0x00`, then `[MPU6500] FAIL`) and is missing from the `SENSORS:` line; the box sends the others' data as normal (section "The serial protocol"). In the application, the connection log under **Settings** → **Advanced** shows the list at every connection: "Sensors it reports: …".
- Change sensors with the box **switched off** (cable out): a sensor plugged in while the box runs is not seen until it restarts.
- For each activity the application picks a connected box that has *all* the sensors it needs. If a box is connected but none has the sensor, the experiment names the missing one, for example "No connected box has the motion sensor plugged in. Switch the box off, plug the motion sensor into one of the three sockets and switch it on again." The banner "No box is sending data." appears only when no box is connected at all.

| Sensor in the socket | The activities that need it |
|---|---|
| The motion sensor (MPU-6500) | Flying Straight, The Box in Balance, The Seismometer, Seismology; the games Among the Clouds, Earthquake!, Resonance, Gyro Target |
| The temperature, humidity and pressure sensor (BME280) | The Magic Thermometer, Pressure and Altitude, Urban Heat Islands (and with Several Boxes), Air Convection, The Water Cycle, Physics of the Atmosphere, Weather Station; together with the light sensor: Solar Panels, Solar Energy |
| The light sensor (VEML7700) | Light is Everywhere, The Beer–Lambert Law, Spectroscopy; Solar Panels and Solar Energy (with the BME280) |
| The $CO_2$ sensor (SCD41) | The Air We Breathe, Indoor Pollution, Air Quality; $CO_2$ and Photosynthesis (with the light sensor); The Greenhouse Effect (with the light sensor and the BME280: all three sockets) |
| The optional sensors (firmware 2.5.0 or newer): the BMV080 particle sensor, the SGP30 or SGP41 vapour sensor, the STEMMA soil probe | Air Before and After Airing (any one of the BMV080, SGP30, SGP41, SCD41); Soil and Water (the soil probe); Air Quality (PM2.5 and the vapours, alongside $CO_2$). These lessons stay hidden until they are ticked in Settings, Visible activities |
| Any | Measurement Notebook |

## 2.2 Microcontroller: ESP32

| Parameter | Value |
|---|---|
| CPU | Xtensa LX6 dual-core, 240 MHz |
| RAM | 520 KB SRAM |
| Flash | 4 MB |
| Interfaces | $I^2C$, SPI, UART, Wi-Fi, BLE |
| $I^2C$ (firmware) | SDA=GPIO21, SCL=GPIO22, 100 kHz |
| Power | 3.3 V |
| USB-serial adapter | CP210x or CH340 (depends on the board) |

**Why 100 kHz?** The $I^2C$ bus runs at 100 kHz because the SCD41 sensor supports a maximum of 100 kHz per the Sensirion datasheet. At 400 kHz, the SCD41 corrupted the other sensors' transactions through *clock stretching*, and the MPU-6500 disappeared from the bus.

**The box warms itself.** The ESP32 dissipates around half a watt (more with Wi-Fi active), and the sensors share its enclosure. The BME280 therefore reads the air inside the box, not the room air: after a few minutes of running, the temperature can read one to a few degrees above the room, and the relative humidity correspondingly lower. To compare boxes, first place them side by side for a few minutes and note how they differ; for absolute values, let the box settle and compare it with a reference thermometer.

## 2.3 IMU sensor: MPU-6500

### 2.3.1 Specifications

| Parameter | Configured value | Notes |
|---|---|---|
| Accelerometer | $\pm 8\,g$ | 4096 LSB/$g$ |
| Gyroscope | $\pm 500\,^\circ/\mathrm{s}$ | 65.5 LSB/(°/s) |
| Magnetometer | — | the MPU-6500 has no magnetometer |
| Low-pass filter (DLPF) | $\approx 20\,\mathrm{Hz}$ | from firmware 2.2.5 |
| Zero offset | $\approx \pm 0.06\,g$ | typical, per axis (datasheet) |
| $I^2C$ address | `0x68` | — |
| Output rate | 50 Hz | $\Delta t = 20$ ms |

The zero offset is not calibrated in the firmware. In practice a box lying flat may show a few hundredths of a $g$ on $a_x$ and $a_y$ (an apparent angle of $1 \ldots 3^\circ$) — which is why the *Flying Straight* exercise uses the 180° reversal test: half the sum of the two readings is the offset, half the difference is the real tilt of the table.

### 2.3.2 LSB-to-physical-units conversion

The firmware reads 14 consecutive bytes starting at register `0x3B` (3 accelerometer axes + internal temperature + 3 gyroscope axes):

**Accelerometer and gyroscope conversion**

$$a_x = s_{ax} \cdot \frac{\text{raw}_{ax}}{4096} \quad [g] \tag{2.1}$$

$$\omega_x = s_{gx} \cdot \frac{\text{raw}_{gx}}{65.5} \quad [^\circ/\text{s}] \tag{2.2}$$

where $s_{ax}$, $s_{gx} \in \{-1, +1\}$ are the mounting-correction signs.

Firmware implementation (C):

```
out[0] = (float)((int16_t)(b[0]<<8|b[1])) / 4096.0f;  // ax
out[3] = (float)((int16_t)(b[8]<<8|b[9])) / 65.5f;    // gx
if (imu_flipped)          // IMU_MOUNT FLIPPED, the default (see below)
  for (int i = 0; i < 6; i++) out[i] *= IMU_FLIP_SIGN[i];  // -1,+1,-1,-1,+1,-1
```

### 2.3.3 The on-chip low-pass filter (DLPF)

The box sends 50 samples per second, so it can faithfully reproduce only signals below the 25 Hz Nyquist frequency. Out of reset, unconfigured, the MPU-6500 passes up to 250 Hz (gyroscope) and 460 Hz (accelerometer): the vibration of a fan or a motor then folds down into fake slow signals (*aliasing*). From version **2.2.5**, the firmware writes at every initialisation:

```
mpuWrite(0x1A, 0x04); // CONFIG:        gyro  DLPF_CFG=4   -> ~20 Hz, ~10 ms delay
mpuWrite(0x1D, 0x04); // ACCEL_CONFIG2: accel A_DLPF_CFG=4 -> ~21 Hz, ~12 ms delay
```

Hand movements, the oscillations in *Resonance* (1 . . . 3.3 Hz) and the taps in *The Seismometer* pass practically unattenuated; components above ≈ 20 Hz are attenuated instead of showing up as fake slow motion. The filter is re-applied after an automatic IMU re-initialisation too.

### 2.3.4 Mounting correction (180° rotation about the Y axis)

The MPU-6500 board is mounted face down inside the box. The firmware corrects this mounting with a **180° rotation about the Y axis** — never by mirroring the Z axis alone, which would invert the reference frame's orientation instead of restoring it to an upright one. A 180° rotation about Y leaves the Y axis unchanged and requires negating the other **two** axes (for the accelerometer and the gyroscope alike):

| Axis | Accelerometer sign | Gyroscope sign |
|---|---|---|
| X | −1 | −1 |
| Y | +1 | +1 |
| Z | −1 | −1 |

As a result, a box lying flat (on its bottom) reports $a_z \approx +1\,g$: the table pushes the box up, and the accelerometer measures that support force (in free fall it would read 0). **A box built so far (sensor face down) that reports $a_z \approx -1\,g$ while lying flat is running old firmware (`serial_wifi` older than `v2.2.1`, or an old USB-only `serial_only`) and must be reflashed with `serial_wifi_v2_5_0`** — the wrong sign is a firmware defect, not something to compensate for in the application. From v5.3 the application shows this warning itself (see Ch. 4).

**A box whose sensor is mounted the other way (the new board).** Every box built so far has the MPU-6500 board face down. A newer board has it face up: with the same firmware such a box shows $a_z \approx -1\,g$ on the table although its firmware is current (2.2.1 or newer). No other firmware is needed; the correction is switched off once for that box.

- **Firmware 2.3.0 and newer keep the mounting in the box**, in its non-volatile memory (NVS namespace `magicbox_hw`): `FLIPPED`, the default and right for every box built so far (readings turned over, exactly as in 2.2.6), or `NORMAL` (sensor face up, readings pass unchanged). The box announces it at start-up in the `IMU_MOUNT:` line and takes the commands `IMU_MOUNT NORMAL` and `IMU_MOUNT FLIPPED` (section 3.2.2). The setting survives power-off, reflashing and the BOOT-button Wi-Fi reset.
- **In the application**, **Settings → Box names**, tick **Motion sensor upside down** for that box. For a box with firmware 2.3.0 or newer the application sends `IMU_MOUNT NORMAL` to the box (untick: `IMU_MOUNT FLIPPED`), so the setting lives in the box and counts on every computer; the row then says "motion sensor setting kept in the box". The box must be connected when you tick: if it does not answer, the tick springs back and the row says "Not changed: the box did not answer. Connect it and try again." The same can be done without the application, by typing `IMU_MOUNT NORMAL` in the Arduino Serial Monitor right after flashing (see `FLASHING_GUIDE`).
- **Older firmware (2.2.1 to 2.2.6)**: the tick stays on this computer, remembered by the box's ID (`BOX_XXXX`). The application then applies the same 180° rotation about Y (negating X and Z, for the accelerometer and the gyroscope) before it shows, saves or uses the data in activities. If such a box is later reflashed with 2.3.0, the tick moves into the box by itself the next time it connects, and the computer stops turning its data, so they are never turned twice.

*Check rule:* tilt the right edge down ⇒ roll and the gyroscope must have the same sign.

### 2.3.5 Computing angles from the accelerometer

**Roll and pitch (from static acceleration)**

$$\phi_{\text{roll}} = \arctan\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right) \cdot \frac{180^\circ}{\pi} \tag{2.3}$$

$$\theta_{\text{pitch}} = \arctan\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right) \cdot \frac{180^\circ}{\pi} \tag{2.4}$$

Input: $a_x, a_y, a_z$ in $g$; output in degrees (°).

**Full roll** (the aircraft picture over ±180°, including upside-down flight):

$$\phi_{\text{full}} = \operatorname{atan2}(a_y,\, a_z) \cdot \frac{180^\circ}{\pi}$$

### 2.3.6 Data emitted (JSON)

IMU packet (50 Hz):

```
{"id":"BOX_A1B2","ts_ms":12345,"imu_hz":50,"imu
    ":[[-0.023,0.014,0.998,0.152,-0.089,0.041]]}
```

Field order: $[a_x, a_y, a_z, \omega_x, \omega_y, \omega_z]$, accelerations in $g$, angular rates in °/s.

## 2.4 Weather sensor: BME280

### 2.4.1 Specifications

| Quantity | Range | Resolution | Accuracy (datasheet) | Measured noise |
|---|---|---|---|---|
| Temperature | $-40 \ldots +85$ °C | 0.01 °C | ±1 °C | 0.003 °C |
| Humidity | 0 … 100 % | 0.01 % | ±3 % RH | 0.1 % RH |
| Pressure | 300 … 1100 hPa | 0.18 Pa | ±1 hPa absolute<br>±0.12 hPa relative | 1.4 Pa |

I²C address `0x76` or `0x77`; normal mode, ×1 oversampling, ×2 IIR filter, one reading per second. *Measured noise* is the standard deviation of successive readings from a box at rest: it shows how small a *change* the box can see. *Accuracy* shows how far the absolute value can be from the true one. The difference matters:

- Pressure: a difference of 0.12 hPa (one metre of altitude) is clearly visible, even though the absolute value may be off by up to 1 hPa. The 1.4 Pa noise corresponds to about 12 cm of altitude.
- Temperature: the box sees changes of hundredths of a degree, but two boxes can differ by a degree or more — partly because of self-heating (see the ESP32 note).

The firmware sends values with **2 decimal places**: with a single decimal, the chart showed 0.1 °C steps and the real resolution was wasted. The application *displays* pressure with one decimal (two at the **Scientist** level) and altitude with one decimal; data files keep every decimal received.

### 2.4.2 The barometric formula (altitude from pressure)

**International barometric formula (standard atmosphere)**

$$h = \frac{T_0}{L}\left[1 - \left(\frac{P}{P_0}\right)^{\frac{R^* L}{g_0 M}}\right] = 44\,330 \cdot \left[1 - \left(\frac{P}{P_0}\right)^{0.1903}\right] \tag{2.5}$$

- $h$ = altitude [m]; $P$ = measured pressure [hPa]
- $P_0$ = sea-level pressure (default 1013.25 hPa; the QNH value is entered in **Settings** and kept)
- $T_0 = 288.15$ K, $L = 0.0065$ K/m (the temperature lapse rate), so $T_0/L \approx 44\,330$ m
- exponent $\dfrac{R^* L}{g_0 M} = \dfrac{8.3145 \times 0.0065}{9.80665 \times 0.028964} \approx 0.1903 \approx \dfrac{1}{5.255}$, with $R^*$ the universal gas constant and $M$ the molar mass of dry air

**Practical sensitivity:** $\Delta P \approx 0.12$ hPa per metre ($\approx 8.3$ m/hPa) near sea level. One floor ($\approx 3$ m) produces $\Delta P \approx 0.35$ hPa.

With $P_0 = 1013.25$ hPa, the altitude shown is not the altitude of the place: atmospheric pressure changes from day to day by tens of hPa, i.e. hundreds of metres of apparent altitude. To compare with your town's altitude, enter the day's QNH (from an airport or a nearby

weather station) in **Settings**; even then an uncertainty of the order of ±10 m remains from the sensor's absolute accuracy.

### 2.4.3 Dew point (Magnus formula)

**Dew point — the Magnus formula**

$$\gamma(T,\,H) = \frac{aT}{b+T} + \ln\!\left(\frac{H}{100}\right) \tag{2.6}$$

$$T_d = \frac{b\,\gamma}{a-\gamma} \tag{2.7}$$

Constants (Alduchov–Eskridge): $a = 17.625$, $b = 243.04\,^{\circ}\mathrm{C}$; input $T$ in °C, $H$ in %. Every screen of the application uses the same constants.

### 2.4.4 Data emitted (JSON)

Slow-sensor packet (1 Hz):

```
{"id":"BOX_A1B2","ts_ms":45123,"data":{"lux":312.5,"white":1840,"raw_als":1821,
 "T":22.34,"H":58.12,"P":1013.28,"ALT":0.3}}
```

## 2.5 Light sensor: VEML7700

### 2.5.1 Specifications

| Parameter | Firmware ≤ 2.2.4 | Firmware ≥ 2.2.5 |
|---|---|---|
| Useful range | 0 … ≈ 3 800 lx (saturates) | ≈ 0.004 lx … direct sun (capped at 150 000 lx) |
| Gain / integration time | fixed: ×1, 100 ms | automatic: $\times\frac{1}{8}$ … × 2, 25 … 800 ms |
| Resolution | 0.0576 lx | 0.0036 … 1.84 lx (per rung) |
| Non-linearity correction | no | yes, within its valid range |
| Accuracy | ≈ ±10 % below ≈ 10 klx; approximate above (not yet checked against a lux meter) | |
| I²C address | `0x10` (fixed) | |
| Output rate | 1 Hz (the last valid reading is sent) | |

**Up to firmware 2.2.4**, the sensor ran at a fixed gain of ×1 and 100 ms integration. Its 16-bit counter filled up at 65 535 × 0.0576 ≈ 3 775 lx: *any* brighter light — near a sunny window, outdoors in the shade, in the sun — showed as ≈ 3 800 lx. A box that reads ≈ 3 800 lx in full sun must be reflashed.

**From firmware 2.2.5**, the firmware chooses gain and integration time itself from a 9-rung ladder (from $\times\frac{1}{8}$/25 ms for direct sun to ×2/800 ms for near darkness). It steps to a less sensitive rung when the count exceeds 50 000 (two rungs at once when saturated) and to a more sensitive one when it drops below 1 000. The switch never pauses the main loop (a state machine that only compares `millis()`), so the 50 Hz IMU stream is not interrupted. The least sensitive rung (1.8432 lx/count) reaches 65 535 × 1.8432 ≈ 120 800 lx before the non-linearity correction, i.e. direct summer sun; after the correction the value sent is capped at 150 000 lx.

After a rung change the first new value arrives after about two integrations; until then the last valid reading is re-sent.

**Absolute values in bright light.** Above ≈ 10 000 lx, absolute values are approximate: they have not yet been checked against a reference lux meter (the bench test is planned). The activities rely on relative comparisons — the panel's tilt series, sun versus shade, the same box in the same place —, which remain sound.

### 2.5.2 Parameters emitted

`LUX` [lx] — illuminance computed from the ALS channel. From 2.2.5: count × the current rung's resolution, with the Vishay non-linearity correction (application note 84323, the same one the Adafruit library's automatic mode applies), used only up to 10 000 lx, where its polynomial is valid; above that the value continues linearly, joined without a jump (capped at 150 000 lx). Up to 2.2.4: count ×0.0576, without correction — the value was *not* "ALS-compensated".
`WHITE` — the white channel. `RAW` — the ALS channel. From 2.2.5 both are reported *on the old scale* (the equivalent at gain ×1 / 100 ms), so old and new files stay comparable; that is why they can exceed 65 535 in bright light.

### 2.5.3 Interpretation thresholds

| Range [lx] | Description |
|---:|---|
| < 1 | Darkness |
| 1 . . . 50 | Low light (night / dimly lit indoors) |
| 50 . . . 500 | Ordinary indoor lighting |
| 500 . . . 2 000 | Overcast day / near a window |
| 2 000 . . . 20 000 | Indirect daylight, outdoors in the shade |
| 20 000 . . . 80 000 | Direct sun (partial, winter, morning) |
| > 80 000 | Strong direct sun |

The last three rows can only be reached with firmware 2.2.5 or newer; with older firmware, everything above ≈ 3 800 lx falls into the same row.

## 2.6 $CO_2$ sensor: SCD41

### 2.6.1 Specifications

| Parameter | Value | |
|---|---|---|
| Specified range | 400 . . . 5 000 ppm | may also report below 400 ppm |
| Accuracy | ±(40 ppm + 5 % of reading) | within the specified range |
| Principle | photoacoustic NDIR (infrared) | |
| Settling | ≈ 30 s | first readings may be unstable |
| Measurement cycle | ≈ 5 s | set by the sensor, not the firmware |
| Self-calibration (ASC) | on | see below |
| $I^2C$ address | `0x62` | fixed (Sensirion) |

The SCD41 completes a measurement every ≈ 5 s, slower than the other sensors (1 Hz). The firmware keeps the last value read and repeats it in every 1 Hz packet so the data stream stays regular — the $CO_2$ values do not, in fact, update on every packet. From v5.3 the application keeps a time axis of its own for $CO_2$ (see Ch. 4), so charts no longer stretch or squeeze the curve.

**Which firmware reads the SCD41.** The firmware (`serial_wifi`) reads it from 2.2.0. A box still running the old USB-only firmware (`serial_only`, now retired) may ignore the sensor even when it is fitted: reflash it with `serial_wifi_v2_5_0` for the $CO_2$ activities. If the SCD41 is absent from the bus, the firmware ignores it and sends the rest of the data normally.

### 2.6.2 Automatic self-calibration (ASC)

An NDIR sensor slowly loses its reference over time. The SCD41 has *Automatic Self-Calibration* (ASC): it assumes that, over a period of operation (about 44 hours at first, then about 156 hours, i.e. a week), the lowest concentration it has seen is fresh outdoor air, $\approx 400$ ppm, and slowly shifts its reference accordingly.

- **How the firmware sets it.** The firmware (`serial_wifi` 2.2.x) starts the sensor with `scd41.begin(Wire)` from the SparkFun SCD4x library. That function enables ASC at *every boot* (default argument `autoCalibrate = true`). The firmware has no other calibration command.
- **When it helps.** A classroom aired well at least once a week (or a box taken outdoors now and then) keeps its calibration by itself.
- **When it hurts.** A box that never sees fresh air (a cupboard, an unventilated room) will "learn" a higher value as 400 ppm and read too low. Conversely, a sealed jar with a plant in the light, in which $CO_2$ stays below 400 ppm for hours, can push the reference up. For long experiments below 400 ppm, disable ASC.

**Disabling ASC.** Because the firmware re-enables ASC at every boot, it is disabled in the box's sketch (`serial_wifi`), by changing a single line and reflashing the box:

```
scd41_ok = scd41.begin(Wire, true, false);   // measBegin=true, autoCalibrate=false
```

**Forced recalibration (FRC) outdoors.** If a box reads visibly wrong in clean air, it can be recalibrated against a known concentration:

1. Take the box outdoors, in the shade, away from people, cars and chimneys, and let it measure for at least 3 minutes (outdoor air is about 420 ppm today).
2. With a short sketch (or calls added temporarily to the firmware), stop periodic measurement and request the recalibration; the function returns the correction applied, in ppm:

   ```
   scd41.stopPeriodicMeasurement();          // waits 500 ms
   float corr = scd41.performForcedRecalibration(420);
   ```

3. Restart periodic measurement (`startPeriodicMeasurement()`) or restart the box, and check that it reads around 420 ppm outdoors.

The current firmware exposes no calibration commands on the serial port; both operations therefore require reflashing the box (see `FLASHING_GUIDE.pdf` or `FLASHING_GUIDE_RO.pdf` in the `firmware` folder). Do not run an FRC in the classroom: the concentration there is not known.

### 2.6.3 Thresholds used by the application (a ventilation indicator)

At classroom levels, $CO_2$ is not a poison: it is an *indicator of ventilation* — how much of the air people have breathed out is still in the room. The WHO has no guideline value for indoor $CO_2$; the thresholds below are practical thresholds used by the application, close to the indoor air quality categories of standard **EN 16798-1** (which expresses them as the excess over outdoor air: roughly +550, +800 and +1350 ppm for categories I, II and III).

| $CO_2$ [ppm] | Band in the application | Action |
|---|---|---|
| $< 800$ | Excellent | — |
| $800 \ldots 1200$ | Good | — |
| $1200 \ldots 2000$ | Moderate | The air is getting stale: ventilate |
| $2000 \ldots 5000$ | Poor | Open the windows now |
| $> 5000$ | Dangerous | Ventilate; the workplace exposure limit is 5000 ppm (8-h average) |

### 2.6.4 $CO_2$ rate of change

**Rate of change = slope of the regression line**

$$\dot{C} = 60 \cdot \frac{\sum_i (t_i - \bar{t})(C_i - \bar{C})}{\sum_i (t_i - \bar{t})^2} \quad \text{[ppm/min]}, \qquad t_i \text{ in seconds} \tag{2.8}$$

The exercises no longer divide the difference of two readings by the time between them (two nearby readings differ by noise and by the sensor's 5 s cycle). They fit a line through every reading in a window and show the slope with its uncertainty (the standard error of the slope, ±): the last 3 minutes in *Indoor Pollution*, the last 5 minutes (but at least 3) in *$CO_2$ and Photosynthesis*. The colours are neutral: a falling $CO_2$ is not "bad", and a rising one is not "good".

### 2.6.5 Data emitted (JSON)


```
{"id":"BOX_C3D4","ts_ms":5000,"data":{"co2":842,"TEMP_SCD":23.1,"HUM_SCD":55.4}}
```

# Chapter 3

# Firmware

## 3.1 Version history

The box has one firmware, `serial_wifi` (2.x), which works over USB and over Wi-Fi and reads every sensor, $CO_2$ included. Until 25 September 2026 a USB-only variant, `serial_only` (1.x, last 1.3.4), advanced in step with it; it is now **retired** and no longer supplied. A box still running it is reflashed with `serial_wifi_v2_5_0`. The table keeps its column for reference.

| Wi-Fi | USB only (retired) | Main changes |
|---|---|---|
| 2.0.0 | 1.2.0 | Dual USB + Wi-Fi transport; IMU mounting correction |
| 2.1.0 | — | Wi-Fi setup portal (access point), no passwords in the code |
| 2.2.0 | — | SCD41 support ($CO_2$ + T/RH), 5 s cycle |
| 2.2.1 | 1.3.0 | T/RH/P with 2 decimals; mounting correction as a rotation about Y (**minimum supported version**: below it, $a_z \approx -1\,g$) |
| 2.2.2 | 1.3.1 | $I^2C$ bus at 100 kHz |
| 2.2.3 | 1.3.2 | IMU self-healing: boot retries + bus reset |
| 2.2.4 | 1.3.3 | Bus unlock with 9 clock pulses; BME/VEML retry at boot; gives up after 30 attempts |
| 2.2.5 | 1.3.4 | VEML7700 auto-range (up to direct sunlight); IMU low-pass filter ≈ 20 Hz; (USB only) SCD41 support; (Wi-Fi) non-blocking TCP connect, with progressively spaced retries |
| 2.2.6 | 1.3.4 | (Wi-Fi only) the setup network is named MagicBox-XXXX, the (English) page lists the 2.4 GHz networks the box sees with their signal, the serial line says why the box could not join, and the box retries the saved network every 2 minutes |
| 2.3.0 | — | **Current.** The box keeps which way up its motion sensor is mounted (`IMU_MOUNT`, kept in the box), for the new board with the sensor face up; `INFO` command. With the default setting the readings are exactly those of 2.2.6 |

## 3.2 The serial protocol

**Baud rate**: 115200, 8N1. The same lines are sent over Wi-Fi.

### 3.2.1 Boot sequence (once)

```
[BOOT] MagicBox fw v2.3.0 (build Sep 26 2026)
[IMU] Sensor mounting: FLIPPED
[VEML7700] OK
[BME280] OK
[MPU6500] OK
[SCD41] OK
DEVICE: BOX_A1B2
TYPE: MULTI
SENSORS: LUX,WHITE,RAW,T,H,P,ALT,AX,AY,AZ,GX,GY,GZ,CO2,TEMP_SCD,HUM_SCD
FW: 2.3.0
IMU_MOUNT: FLIPPED
```

The application parses `DEVICE:`, `TYPE:`, `SENSORS:`, `FW:` and `IMU_MOUNT:`. Lines starting with `[` (diagnostics) are ignored. The header is re-sent over TCP on every new connection. `IMU_MOUNT:` exists from firmware 2.3.0 on: `FLIPPED` (the default, every box built so far) or `NORMAL` (a board with the motion sensor face up); an older box does not send it.

The sequence above is that of a box with all four sensors, as an example. In practice the box has three sockets (section 2.1), and the sensor that is not fitted shows `FAIL` and is missing from `SENSORS:`. A box without the MPU-6500, for example:

```
[MPU6500] WHO_AM_I=0x00
[MPU6500] FAIL
SENSORS: LUX,WHITE,RAW,T,H,P,ALT,CO2,TEMP_SCD,HUM_SCD
```

This is normal behaviour, not a fault; a box without the MPU-6500 sends no `"imu"` packets either.

### 3.2.2 Commands to the box (firmware 2.3.0 and newer)

The box also listens. A command is one line, sent over USB or over the Wi-Fi link; the Arduino Serial Monitor at 115200 baud works too, with any line-ending setting. Upper or lower case does not matter.

| Command | Effect and reply |
|---|---|
| `IMU_MOUNT NORMAL` | The motion sensor sits face up (the new board): readings pass unchanged. Reply: `[IMU] Sensor mounting saved: NORMAL (readings unchanged)`, then `IMU_MOUNT: NORMAL` |
| `IMU_MOUNT FLIPPED` | The sensor sits face down (every box built so far, the default): readings are turned over. Reply: `[IMU] Sensor mounting saved: FLIPPED (readings turned over)`, then `IMU_MOUNT: FLIPPED` |
| `INFO` | Prints the header lines again (`DEVICE:` to `IMU_MOUNT:`) |
| anything else | `[CMD] Unknown command: X` |

The mounting is saved in the box (NVS namespace `magicbox_hw`), separately from the Wi-Fi settings: it survives power-off and reflashing, and the BOOT-button Wi-Fi reset does not erase it. The application uses these commands for the **Motion sensor upside down** tick (section 2.3.4).

### 3.2.3 Main loop — timing

| Event | Frequency | JSON key |
|---|---|---|
| Slow sensors (VEML7700, BME280; SCD41 held) | 1 Hz | `"data"` |
| IMU (MPU-6500) | 50 Hz | `"imu"` |
| SCD41 read | 0.2 Hz | included in `"data"` |

## 3.3 The device ID

**Generating the ID from the MAC address (C)**

```
snprintf(DEVICE_ID, sizeof(DEVICE_ID), "BOX_%04X",
         (uint16_t)(ESP.getEfuseMac() >> 32));
```

Bytes 4–5 of the 6-byte MAC (eFuse) ⇒ the format `BOX_XXXX`.
Example: MAC `AA:BB:CC:DD:EE:FF` ⇒ `BOX_DDEE`.

In the application, the teacher can give each box a name (for example "Group 1") in **Settings → Box names**; the name is kept and appears everywhere in place of the ID, together with a fixed colour per box. The ID remains in the data files.

## 3.4 I²C bus self-healing — the 9-clock method

The SCD41 can hold SDA low (*clock stretching*) if power is lost mid-measurement. `Wire.end()` alone does not release the bus.

**Unlocking procedure (9 clock pulses)**

1. Set SDA (GPIO 21) as an input with pull-up
2. Generate 9 manual pulses on SCL (GPIO 22): HIGH / LOW, 5 µs each
3. Stop at the first cycle where SDA goes HIGH (the device has released the bus)
4. Generate a STOP condition: SDA low→high with SCL=HIGH
5. `Wire.end()` + `Wire.begin(21,22)` + `Wire.setClock(100000)`
6. Re-initialise the MPU-6500 (including the low-pass filter)
7. After 30 failed attempts: IMU disabled until the next boot (BME280 + SCD41 continue normally)

## 3.5 Wi-Fi: setup, discovery, reconnection

`serial_wifi` firmware (2.x) only.

- **Setup.** With no saved network, the box opens an open access point. Its name is `MagicBox-XXXX`, where XXXX are the last 4 characters of the box ID (`CutiaMagica-XXXX` on firmware older than 2.2.6). Connect to it with a phone or laptop. On some phones the setup page opens by itself. On others (Samsung, for example) it does not: the phone says *"Internet may not be available"*; tap *"Connect only this time"*, then open a browser at `http://192.168.4.1`. The setup page is in English: choose the classroom network under *"WiFi network"* (or type its name under *"…or type the network name"*), enter the *"Password"* (*"Show password"* helps when typing on a phone) and tap *"Save and connect"*. The page shows *"Saved!"*, the box restarts and connects by itself from then on. If the saved network does not answer within 20 s, the portal reopens.

- **When the box cannot join** (firmware 2.2.6 or newer). The box sees **2.4 GHz networks only** (the page says so: *"The box sees 2.4 GHz networks only."*), so a dual-band router must keep its 2.4 GHz band on. The list on the page shows the 2.4 GHz networks the box sees, strongest first, with dots for the signal and a padlock for those with a password. The page does not say why a join failed; the reason is printed only on the serial line, e.g. `[WiFi] Cannot join "X": network not found (out of range, or 5 GHz only) (reason 201)` or `wrong password`, followed by the list of networks seen. While nobody is connected to the page, the box tries the saved network again every 2 minutes, so a box switched on before the router joins by itself once the router is up.
- **Erasing the saved network.** Hold the BOOT button while powering the box.
- **Discovery.** Every 3 s the application sends a UDP *broadcast* message on port 8766, of the form `MAGICBOX:EDUGEOLAB:<ip>:<port>`. The box listens for it and opens a TCP connection to the computer on port **8765**. The `MAGICBOX:` prefix and the `EDUGEOLAB` token must match in the application and the firmware; they are never renamed.
- **Spaced-out reconnection.** If the computer's firewall blocks port 8765, the connection fails. From 2.2.5 the connection attempt no longer blocks the loop (until then each attempt stopped the IMU stream for about 1.5 s every 5 s), and the pause between attempts doubles: $5 \rightarrow 10 \rightarrow 20 \rightarrow 40 \rightarrow 60$ s (maximum). It resets on a successful connection or when the discovery message comes from a different computer.
- **Dual transport.** The box sends the same lines over USB and Wi-Fi; the application drops packets received twice by their timestamp.
- **Network limits.** School networks with client isolation (devices cannot see each other) or WPA2-Enterprise authentication do not work; the fallback is a phone hotspot, joined by both the boxes and the computer.

# Chapter 4

# Data flow in the application

## 4.1 SerialManager

`SerialManager` scans COM ports every 3 s and launches one `SerialWorker` (QThread) per port. Each worker:

1. Opens the port at 115200 baud
2. Reads ASCII lines, identifies the device (at most 4 s) or infers the type from the first JSON packet's keys (if the box was already running)
3. Emits the `identified(device_id, type, sensors, port)` signal
4. Continues emitting `data_ready(device_id, payload)`
5. No data for 8 s ⇒ automatic reconnection

Over Wi-Fi, `WifiManager` does the same: it listens on TCP 8765 and sends the discovery messages.

### 4.1.1 Inferring the type from JSON keys

| Keys present in `"data"` | Detected type |
|---|---|
| `T, H, P, ALT` | BME280 |
| `lux, white, raw_als` | VEML7700 |
| `co2, TEMP_SCD, HUM_SCD` | SCD41 |
| `"imu"` packets (50 Hz, no `"data"`) | MPU6500 |
| Combination | MULTI |

### 4.1.2 Firmware version

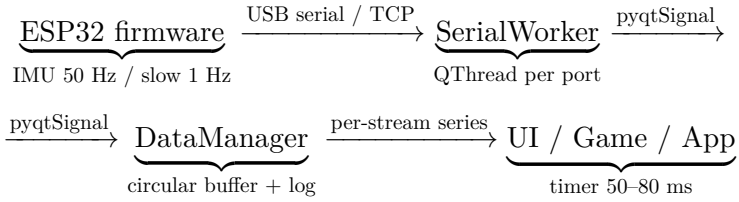
The application reads the `FW: x.y.z` line. Below **2.2.1** it shows a warning that the box must be reflashed (the Z axis would be upside down); a box running the old USB-only firmware (1.x) is told to flash `serial_wifi` 2.3.0 instead, since it works over USB too. From 2.2.1 to 2.2.5 it shows only a note that version 2.3.0 exists (below 2.2.5, one in which the light sensor reaches full sunlight). From **2.2.6** on it says nothing: 2.3.0 only adds the sensor mounting kept in the box, which the boxes already in schools do not need, so a 2.2.6 box does not have to be reflashed.

## 4.2 DataManager

Keeps a circular window per box in memory and writes everything to disk.

- **A separate time axis per stream.** The IMU (50 Hz), the slow sensors (1 Hz) and $CO_2$ (every 5 s) each keep their own timestamps. Up to v5.2 all streams shared a single list of times, and a 10-minute $CO_2$ curve could be drawn over 12 seconds.
- `get_latest(dev_id)` — the most recent point per sensor
- Time series are read per stream, with their own times; exercises and apps pick the box among those connected at that moment.
- Parsing: each handler (`MPU6500Handler`, `BME280Handler` etc.) maps the raw JSON keys to the standard keys (`AX`, `T`, `LUX` etc.)

## 4.3 Full data path




Typical USB-serial → chart latency: **50–100 ms** (plus ≈ 10 ms added by the IMU low-pass filter).

## 4.4 Logs, exports and settings (v5.4)

### 4.4.1 Logs on disk

- While **Save every box's data to a file automatically (recommended)** under **Settings → Data recording** is ticked (the default), each box has, in the `logs` folder next to the application, one file for the slow sensors and one for the IMU:
  `<ID>_<YYYYMMDD>_<hhmmss>.csv`,
  `<ID>_<YYYYMMDD>_<hhmmss>_imu.csv`.
- **Fallback folder.** If the application folder cannot be written to (a write-protected stick, a network share, `Program Files`), the application writes to `MagicBox\logs` in `Documents` (as a last resort, in the temporary folder) and shows a warning. A write error (for example a file held open in Excel) stops only that box's log, not the charts; the **Data & Export** page shows the real folder.
- **Time.** The first two columns are the date and time, with milliseconds (for example `2026-09-23 10:15:02.345`), and the time elapsed, in seconds, since the box connected. Their headers follow the application's language: `Timestamp`, `Elapsed (s)` (in Romanian `Data și ora`, `Timp scurs (s)`). The elapsed column is the one to chart with in Excel.
- Values in the logs are written in the "machine" format (comma separator, decimal point), whatever the language; only exports adapt to the language.
- **Nothing is deleted without asking.** The **Keep logs (0 = always)** setting defaults to *always*; if the teacher picks a number of days, the application asks at start-up before deleting older logs.

### 4.4.2 Exports

- The **Data & Export** page (visible at the **Scientist** and **Teacher** levels) exports one box as **.xlsx** (the default) or CSV. The Excel file has a legend sheet listing each column, its unit and the sensor's accuracy.
- The CSV export can be chosen *for Excel in Romanian* (**;** separator and decimal comma), so that Romanian Excel opens it in columns; with the application in Romanian this is the default CSV.
- **Export all boxes (Excel)** creates, in the folder you choose, a folder holding one Excel file per box — `MagicBox_<YYYYMMDD>_<hhmm>` —: the whole lesson in one place.

### 4.4.3 Box names and the upside-down motion sensor

Under **Settings** → **Box names**, every box that has connected at least once has a row: its ID, its state (connected, firmware version), a field for a name (for example "Group 1") and the **Motion sensor upside down** tick. The name and the tick are kept per box ID. The name appears on every screen, with a colour and a line style fixed for that box; the ID stays in the log file names. For a box with firmware 2.3.0 or newer the tick is sent to the box and kept there, so it counts on every computer (the row then says "motion sensor setting kept in the box"); the box must be connected when you tick it (section 2.3.4).

### 4.4.4 Settings

- **Age level** and **Visible activities**: which activities the class sees.
- **Box names** (above).
- **Language and display**: **Language:** English (the default), Romanian, French or German, applied at restart; **Projector mode** — bigger chart text and thicker lines, for the charts opened from then on.
- **Boxes over WiFi**: the computer's address and the TCP port the application listens on (8765), useful when troubleshooting the network.
- **Data recording**: automatic saving and **Keep logs (0 = always)**.
- **Altitude Calibration**: **Sea-level pressure (QNH)**, kept between sessions, and **Altitude of the school**, from which the Weather Station works out the pressure reduced to sea level.
- **Advanced** (closed by default): **Rescan USB ports**, **Live chart refresh interval:** and the technical connection log. The serial port speed is not a setting: it is 115200 baud, fixed by the firmware.

# Chapter 5

# Educational games

## 5.1 Among the Clouds — a flying game

### 5.1.1 Principle

The box is an aircraft's control stick: roll → left/right, pitch → up/down. Clouds are obstacles; rings score points. Turn the box over by 180° and the plane flies upside down.

### 5.1.2 Control formulas

**Aircraft control from the accelerometer**

$$\phi_{\text{roll}} = \arctan\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right) \tag{5.1}$$

$$\theta_{\text{pitch}} = \arctan\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right) \tag{5.2}$$

$$v_x = \text{clamp}\left(\frac{\phi}{30^\circ}, -1, 1\right) \times 260 \text{ px/s} \tag{5.3}$$

$$v_y = -\text{clamp}\left(\frac{\theta}{30^\circ}, -1, 1\right) \times 260 \text{ px/s} \tag{5.4}$$

Picture rotation (with upside-down flight): $\phi_{\text{full}} = \text{atan2}(a_y, a_z)$

### 5.1.3 Collision detection

$$d = \sqrt{(x_p - x_o)^2 + (y_p - y_o)^2} \tag{5.5}$$

- Cloud hit: $d < r_{\text{cloud}} + 16$
- Passing through a ring: $|x_p - x_o| < 12$ **and** $|y_p - y_o| < r_{\text{ring}} - 14$

### 5.1.4 Game parameters

| Parameter | Value |
|---|---|
| Frames per second | 20 (50 ms tick) |
| World speed (clouds) | 150 px/s; +15 % per 5 rings (max. 330) |
| Max aircraft speed | 260 px/s |
| Angle for max speed | 30° |
| Lives | 3 |
| Protection after a hit | 1.5 s |

In the corner the screen shows the score, the best score (kept between sessions), the lives, the roll and pitch in degrees, live, and the speed factor ("Speed ×1.30"); each time the game speeds up, "Faster!" shows briefly.

## 5.2 Earthquake! — evacuating a building

### 5.2.1 Physical principle

The game simulates how a real earthquake unfolds and tackles the myths about taking shelter:

1. **P wave** (compressional, fast) — the early warning. For the intermediate-depth Vrancea earthquakes (60–200 km), the INFP warning system gives Bucharest about 25 s.
2. **S wave** (shear) — the main shaking; mobility drops sharply
3. **Aftershocks** — calm periods + smaller shocks

### 5.2.2 Shock intensity from magnitude

The magnitude in the game is a moment magnitude ($M_w$), not "on the Richter scale".

**Normalized intensity and time envelope**

$$I(M) = \mathrm{clamp}\left(\frac{M - 3.5}{3.0},\ 0.15,\ 1.25\right) \tag{5.6}$$

$$\mathrm{env}(t) = 0.45 + 0.55 \cdot \frac{t_{\mathrm{remaining}}}{t_{\mathrm{dur}}} \tag{5.7}$$

$$I_{\mathrm{total}} = I(M) \cdot \mathrm{env}(t) \tag{5.8}$$

Shock duration: $t_{\mathrm{dur}} = 2.5 + (M - 4.0) \times 0.9$ [s]

### 5.2.3 Mobility reduction during the shock

$$\Delta x = v_{\mathrm{player}} \cdot \underbrace{\left(1 - 0.75 \cdot \min(I_{\mathrm{total}},\ 1)\right)}_{\text{hindrance factor}} \cdot \Delta t \tag{5.9}$$

At maximum intensity ($I = 1$), the player moves at 25% of normal speed — the realistic effect of a strong earthquake.

### 5.2.4 Shelters and scores

| Shelter type | Radius [px] | Multiplier | Lesson |
|---|---|---|---|
| Under the table (`masa`) | 48 | ×2.0 | Full protection |
| Wall corner (`colt`) | 30 | ×1.5 | Good |
| Door frame (`toc`) | 20 | ×1.2 | Weak (a popular myth!) |
| No shelter | — | ×1.0 | Dangerous |

### 5.2.5 Jump detection (EMA filter on $a_z$)

**A jump = a dynamic spike in $a_z$**

$$\tilde{a}_z[n] = \tilde{a}_z[n-1] + 0.08 \cdot \big(a_z[n] - \tilde{a}_z[n-1]\big) \quad \text{(slow EMA = gravity)} \tag{5.10}$$

$$\delta a_z[n] = a_z[n] - \tilde{a}_z[n] \quad \text{(dynamic component)} \tag{5.11}$$

$$\text{jump} = \delta a_z > \theta_{\text{jump}} \quad \theta_{\text{jump}} \in [0.20,\, 0.80]\, g \text{ (UI slider)} \tag{5.12}$$

### 5.2.6 Aftershock magnitude

$$M_{\text{aftershock}} = \min(M_0 - 0.8,\; M_0 - 2.0 + 0.18 \cdot n_{\text{aftershocks}} + \mathcal{U}(0,\, 0.5)) \tag{5.13}$$

with $M_0 = 5.8$ the mainshock magnitude. The $M_0 - 0.8$ ceiling follows Båth's law: the largest aftershock sits on average about one magnitude unit below the mainshock. (The ceiling used to be 7.0, so aftershocks could exceed the mainshock — unrealistic.)

## 5.3 Resonance — damped harmonic oscillators

### 5.3.1 The physical model

Each building is a **damped harmonic oscillator excited at its base** (a standard model in earthquake engineering):

**Equation of motion with base excitation**

$$\ddot{x} + 2\zeta\omega_0\,\dot{x} + \omega_0^2\,x = -a(t) \tag{5.14}$$

- $x$ = the building's displacement relative to its base [m]
- $\omega_0 = 2\pi f_0$ = the natural angular frequency [rad/s]
- $\zeta = 0.05$ = the damping ratio (5%, typical for civil structures)
- $a(t)$ = the base acceleration (the hand) [m/s$^2$]

Integration by the **semi-implicit (symplectic) Euler** method at each IMU sample ($\Delta t \approx 20$ ms): the velocity is updated first, and the position uses the *new* velocity.

$$\dot{x}[n+1] = \dot{x}[n] + \big(-2\zeta\omega_0\dot{x}[n] - \omega_0^2 x[n] - a[n]\big)\,\Delta t \tag{5.15}$$

$$x[n+1] = x[n] + \dot{x}[n+1]\,\Delta t \tag{5.16}$$

Unlike the explicit Euler method (which would use $\dot{x}[n]$ in the second equation and add spurious energy at every step), the symplectic scheme conserves an undamped oscillator's energy on average and stays stable as long as $\omega_0\Delta t < 2$; for the 3.3 Hz house, $\omega_0\Delta t \approx 0.41$.

### 5.3.2 Fair amplitude normalization

**Normalized amplitude (frequency-independent)**

$$A_{\text{norm}} = |x| \cdot \frac{2\zeta}{X_{\text{ref}}} \tag{5.17}$$

$X_{\text{ref}} = 0.02\,\text{m}$ = the hand's reference displacement.

At resonance: $A_{\text{norm}} \approx 1$ for any building, given the same hand motion.

**Why normalize on displacement, not acceleration?**
The same hand motion (±2 cm) produces $(2\pi f)^2$ times more acceleration at high frequencies. Without normalization, the house (3.3 Hz) would dance at any quick tremor, while the tower (1 Hz) would seem impossible to set in motion.

### 5.3.3 Shake frequency (from zero crossings)

**Frequency estimate from zero crossings**

$$f_{\text{shake}} = \frac{N_{\text{crossings}}}{2 \cdot T_{\text{window}}} \tag{5.18}$$

The signal is $a_y$ in $\text{m/s}^2$, without gravity and tilt (a high-pass filter with $\tau = 1$ s), taken sample by sample by its timestamp. Window: 1.5 s; a crossing counts only if $|a| > 0.4\,\text{m/s}^2$ (filters out noise).

### 5.3.4 Building parameters

| Building | $f_0$ [Hz] | $T_0$ [s] | Floors | Rule of thumb |
|---|---|---|---|---|
| The Tower | 1.0 | 1.0 | 10 | $T \approx N_{\text{floors}}/10$ s |
| The Block | 2.0 | 0.5 | 5 | |
| The House | 3.3 | 0.3 | 3 | |

The buildings follow the rule of thumb $T \approx N/10$ s ($f_0 = 10/N$ Hz). A building wins its star when its normalized amplitude stays above 0.55 for 1.2 s *and* is at least 1.6 times that of either of the other two; hence the on-screen prompt: "Make each building dance on its own".

Deformation shape (cantilever mode): floor $k$'s displacement $\propto (k/N_{\text{floors}})^2$.

## 5.4 Gyro Target — neuroplasticity

Four levels: **Level 1: Steady Center**, **Level 2: Moving Target**, **Level 3: Inverted Controls**, **Level 4: Inverted & Moving**. The player's dot follows roll and pitch (exponentially smoothed, $\alpha = 0.2$); the edge of the arena is 40°, and the five rings are labelled 8°, 16°, …, 40°. A level is passed when the dot stays on the target for 2 s.

### 5.4.1 The target's Lissajous trajectory (levels 2 and 4)

**Lissajous curve**

$$x_{\text{target}}(t) = 0.54\,R \cdot \cos(0.37\,t) \tag{5.19}$$
$$y_{\text{target}}(t) = 0.42\,R \cdot \sin(0.63\,t) \tag{5.20}$$

$R$ = the arena radius, $t$ in seconds. The two angular frequencies are not in a simple ratio, so the target's path does not soon repeat.

### 5.4.2 Reversed controls (levels 3 and 4)

The signs of roll and pitch are flipped in the dot's position (equivalent to $a_x \to -a_x$, $a_y \to -a_y$). The brain has built "up = up" reflexes over years, and knowing the rule does not switch them off. Adaptation *begins* within minutes and shows in the falling times over successive attempts; how long a full switch takes depends on the person and the practice. (The app used to print a figure — "3–7 days of daily practice" — which was not a measured constant and has been removed.)

# Chapter 6

# Guided exercises

## 6.1 The Seismometer

### 6.1.1 Vibration filter (high-pass on each axis)

**Extracting vibration: a separate baseline on each axis**

$$b_k[n] = b_k[n-1] + \alpha\,\big(a_k[n] - b_k[n-1]\big), \quad k \in \{x, y, z\} \tag{6.1}$$

$$\big(\alpha = \tfrac{\Delta t}{\tau + \Delta t},\ \tau = 2\,\mathrm{s};\ \text{at 50 Hz } \alpha \approx 0.01,\ f_c \approx 0.08\,\mathrm{Hz}\big)$$

$$v[n] = \big|\vec{a}[n] - \vec{b}[n]\big| = \sqrt{\textstyle\sum_k \big(a_k[n] - b_k[n]\big)^2} \quad [g] \tag{6.2}$$

$\vec{b}$ is the gravity vector (plus any slow tilt) that the filter tracks; $v$ is the size of the vibration, without gravity, in any direction. On the seismogram, $v$ takes the sign of the axis that deviates most. The filter needs about 3 s of calm to learn the baseline (the calibration step).

**Why per axis, not on the magnitude of the acceleration?** The old version first computed the magnitude $|\vec{a}|$ and then subtracted the baseline. For a small vibration $\vec{\delta}$ on top of gravity, $|\vec{g} + \vec{\delta}| \approx g + \vec{\delta} \cdot \hat{g}$: the magnitude sees only the vertical component, and a sideways tap on the table all but vanishes. Subtracting the baseline on each axis shows the vibration in any direction. Every 50 Hz sample is processed by its own timestamp, not just the latest one received.

### 6.1.2 The shaking index

**Shaking index — a relative logarithmic scale**

$$\text{index} = \log_{10}(PGA_{\text{gal}}) \tag{6.3}$$

$PGA_{\text{gal}} = v_{\text{peak}} \times 980.665$ [cm/s$^2$, Gal].

Orders of magnitude: tapping the table → 0.5 . . . 1.5; a strong impact → 2 . . . 3. Below the calm threshold the index is 0: there it would describe only the sensor's noise.

**This is not a magnitude.** The formula has no distance correction, and the distance correction is exactly what turns a measured amplitude into a magnitude. What it measures is the shaking at the sensor — a quantity akin to intensity. The card used to be called `Richter (ML est.)`; it is now **Shaking index**.

### 6.1.3 Chart scale and status thresholds

The seismogram shows the last 10 s on a fixed scale of $\pm 0.25\,g$, so small shocks stay visible and two recordings can be compared by eye. The measured noise of a box at rest, after the filter, is about $\pm 0.01\,g$; the calm threshold sits just above it (the green band on the chart).

| Peak $v$ [g] | Status shown |
|---|---|
| $< 0.012$ | Calm (within the sensor noise) |
| $0.012 \ldots 0.03$ | Light vibration |
| $0.03 \ldots 0.10$ | Moderate vibration |
| $\geq 0.10$ | Strong shake / impact |

The **Peak amplitude** is the largest $|v|$, held for 3 s and then left to decay; the **Save the peak** button records it. An *event* is an excursion above $0.03\,g$, separated from the previous one by at least 0.5 s of calm.

## 6.2 $CO_2$ and Photosynthesis

### 6.2.1 Chemical context

$$\text{Photosynthesis:} \quad 6\,CO_2 + 6\,H_2O \xrightarrow{h\nu} C_6H_{12}O_6 + 6\,O_2 \tag{6.4}$$

$$\text{Respiration:} \quad C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O + \text{energy} \tag{6.5}$$

### 6.2.2 The rate of change, with its uncertainty

Each step's $CO_2$ rate is the slope of the regression line through the readings of the last 5 minutes of the step (SCD41 section), shown as "$\dot{C} \pm s$ ppm/min". Until 3 minutes of data have been gathered, the application only says how much it has collected and asks for the light to stay unchanged. The capture button records the rate together with the mean illuminance, for the series of light levels.

- Below 200 ppm, $CO_2$ itself limits the plant: the application asks for the jar to be opened for a minute and the step restarted.
- Above 90 % relative humidity, it asks for the jar to be opened and its walls wiped (condensation) before the next step.
- The **light compensation point** (the illuminance at which the rate crosses zero) is found by linear interpolation between the two neighbouring light levels whose rates have opposite signs.

### 6.2.3 Net and gross photosynthesis

In the light, the jar shows *net photosynthesis*: $CO_2$ falls by the difference between what the plant fixes and what it respires ($R_{\text{light}} < 0$). In the dark only respiration shows ($R_{\text{dark}} > 0$). With the slope $R_{\text{soil}}$ of a control (a jar with soil, no plant — moist soil respires too), the application computes:

$$\text{net} = -(R_{\text{light}} - R_{\text{soil}}), \qquad \text{respiration} = R_{\text{dark}} - R_{\text{soil}} \tag{6.6}$$

$$\text{gross} = \text{net} + \text{respiration} = |R_{\text{light}}| + |R_{\text{dark}}| \text{ (for the usual signs)} \tag{6.7}$$

Respiration continues during the day too, hence gross = net + respiration. Measure every rate in the same jar, with the same plant.

### 6.2.4 Mass of $CO_2$ fixed (advanced calculation)

**Mass flux from the concentration change (ideal gas)**

$$\dot{m}_{CO_2}\ [\mathrm{g/min}] = \dot{C}\ [\mathrm{ppm/min}] \times 10^{-6} \times \frac{P\ [\mathrm{Pa}] \times V\ [\mathrm{m^3}]}{R \times T\ [\mathrm{K}]} \times 44.01\ [\mathrm{g/mol}] \tag{6.8}$$

$R = 8.314\ \mathrm{J/(mol\,K)}$; $V$ = the volume of air in the enclosed space, **in m³** (1 L = $10^{-3}$ $\mathrm{m^3}$); $P$ in pascals (1 hPa = 100 Pa); $10^{-6}$ turns ppm into a mole fraction. The factor $PV/RT$ is the total number of moles of air in the container.

**Example:** a 5 L jar ($V = 0.005\,\mathrm{m^3}$), $P = 101\,325$ Pa, $T = 293$ K, $\dot{C} = -20$ ppm/min: $n_{\mathrm{air}} = \frac{101\,325 \times 0.005}{8.314 \times 293} \approx 0.208$ mol; $\dot{m} \approx 20 \times 10^{-6} \times 0.208 \times 44.01 \approx 1.8 \times 10^{-4}$ g/min = 0.18 mg/min fixed. The application shows the same quantity in mg/h (×60: about 11 mg/h), with the jar volume the pupils enter in litres and $P$, $T$ read by the box.

(The previous edition asked for $V$ in litres together with $P$ in pascals, which gave a result 1000 times too large.)

Below 400 ppm the sensor leaves the range in which its accuracy is guaranteed, and hours spent there can influence ASC self-calibration (see the SCD41 section).

## 6.3 Pressure and Altitude

The barometric formula (Ch. 2) applied to the school building: going up, pressure falls by ≈ 0.12 hPa per metre, i.e. by ≈ 0.35 hPa per floor.

- The exercise table has 5 rows: the ground floor, the floors climbed, and **Ground floor (return)**. Returning to the ground floor shows how much the atmospheric pressure changed during the measurement (weather typically changes it by 0.1 . . . 0.3 hPa/h); the difference can be subtracted from the other readings.
- On each floor, wait 1–2 minutes before reading: the sensor and the air in the box need time to settle, and the average steadies. The **Save the value** button fills the current row of the table.
- Pressure is shown with one decimal (two at **Scientist**), and the pressure chart spans at least 1 hPa, so noise does not look like a change.

Chapter 7

# Explore tools

## 7.1 Air Quality — several boxes

Several boxes with an SCD41, each in a different spot in the school. Each box has its own colour and name on every chart; a disconnected box is drawn dashed, marked "(offline)". The coloured background bands are the thresholds of the SCD41 section: **Excellent** ($< 800$), **Good** ($800 \ldots 1200$), **Moderate** ($1200 \ldots 2000$), **Poor** ($2000 \ldots 5000$), **Dangerous** ($> 5000$ ppm).

**The verdict** has two lines, which answer different questions:

- *The level*, from the highest value $C_{\max}$: from 2000 ppm, "open the windows now"; from 1200 ppm, "the air is getting stale"; otherwise all places are well ventilated.
- *The difference* $\Delta = C_{\max} - C_{\min}$ (from two boxes up): $\Delta \geq 600$ ppm names the most loaded place; $\Delta \geq 300$ ppm says the places differ; below 300 ppm the application first asks for the boxes to be put side by side, because two SCD41 sensors in the same air can differ by up to about 180 ppm.

A useful order of magnitude for the classroom: 30 pupils in a room of about 170 $m^3$, windows closed, raise $CO_2$ by roughly 1000 ppm in 30 minutes.

## 7.2 Heat Islands with Several Boxes

Same architecture, with the BME280: compares the temperature in the corners of the classroom, in the hallway and outdoors. Before comparing, keep the boxes side by side for a few minutes so the differences between the sensors are known. Outdoors, a box without a ventilated shelter, left in the sun, also measures the radiation heating it, not just the air: "sun versus shade" differences of several degrees are mostly radiation error. The urban heat island proper is the difference in *air* temperature between the town and its surroundings, and it is largest at night.

## 7.3 Data export (PNG and Excel)

- **PNG**: saves every chart of the current step.
- **Excel (openpyxl)** from an exercise step: a metadata header (title, step, date, box), then the *raw* (unfiltered) readings of that step, with the `elapsed_s` column (seconds since the start of the step). The *Journal* sheet keeps the predictions and observations the pupils typed.
- The pupils' journal is saved as they type, as a text file in the `logs` folder: `<exercise>_<box>_<YYYY-MM-DD_hhmm>_journal.txt`

(the activity ID and the box name). At **Finish** an Excel file with the same name (`.xlsx`) is written beside it: the *Journal* sheet and one sheet of measurements per step. The end screen says exactly what was saved and where.

- The **Measurement Notebook** exports the saved points as `.xlsx` (the default) or CSV, with five columns: `label`, `quantity`, `value`, `unit`, `time` (the label, the quantity key — `T`, `LUX` and so on —, the value, the unit, the time). Each point is also appended at once to the file
  `logs/notebook_<box>_<YYYY-MM-DD>.csv`
  (columns `time`, `label`, `quantity`, `value`, `unit`), as a safety net if the class forgets to export.

*Pedagogical rationale:* an exercise stopped at step 2 of 6 still produces useful data — it can be compared with other classes or with future measurements.

# Chapter 8

# Windows: installation and troubleshooting

## 8.1 Installing the package

- **Where to unzip.** Unzip the package into `Documents` or onto a stick, **not** into `Program Files`: the application may not write there, and the logs would end up in the fallback folder `MagicBox\logs` in `Documents`.
- **SmartScreen.** On first start, Windows may show "Windows protected your PC", because the application is not digitally signed. Click **More info** → **Run anyway**.
- **Smart App Control.** On Windows 11 it can block the unsigned `MagicBox.exe` outright (no *Run anyway*). Switch it off once, as administrator: Windows Security → App & browser control → Smart App Control settings → Off. The antivirus stays on. The user manual shows the screens.
- **The manuals** are in `MagicBox\docs`, next to the application; each activity's **Open guide** button opens the matching PDF.

## 8.2 USB and Wi-Fi connection

- **USB drivers.** ESP32 boards use a CP210x (Silicon Labs) or CH340 (WCH) USB-serial adapter. If a box connected by USB does not appear in the application, nor in *Device Manager* under *Ports (COM & LPT)*, install the adapter maker's driver. Use a USB data cable, not a charge-only one.
- **Firewall.** On first start, Windows asks whether the application may communicate on the network. Tick **Private networks** and allow access: the boxes connect to the computer on **TCP 8765**, after the application announces itself by **UDP** broadcast (port 8766). The classroom Wi-Fi must be marked as *private* in Windows; on a *public* network the firewall blocks the connections.
- **A box does not appear over Wi-Fi.** Check, in order: the firewall (above); that box and computer are on the same network; that the network does not isolate clients or require WPA2-Enterprise authentication; as a fallback, a phone hotspot. A box that fails to connect retries by itself at most every 60 s.

## 8.3 Common problems

| Symptom | Remedy |
|---|---|
| At power-on a sensor shows `FAIL` (for example `[MPU6500] WHO_AM_I=0x00`) and is missing from `SENSORS:` | Normal: the box has three sockets, so one of the four sensors is always left out. If the activity needs it, switch the box off, plug it in, restart. |
| The activity shows "No connected box has the … sensor plugged in" | The box does not have the sensor the activity needs in a socket (for example the motion sensor for the games, or the $CO_2$ sensor for Air Quality): switch it off, plug the sensor in, switch it on; see the table in section 2.1. |
| The activity shows "No box is sending data." | No box is connected: check the USB cable or the WiFi network. |
| A box lying on the table shows $a_z \approx -1\,g$; the plane flies backwards | Firmware older than 2.2.1 — reflash the box (the application shows the warning). With current firmware: the sensor is mounted the other way — tick **Motion sensor upside down** for that box in **Settings** → **Box names** (with firmware 2.3.0 or newer, while the box is connected: the setting is then kept in the box). |
| Light stops at ≈ 3 800 lx in the sun | Firmware older than 2.2.5 — reflash the box. |
| No $CO_2$ values | The SCD41 is not in a socket; or a box with the old USB-only firmware: reflash it with `serial_wifi_v2_5_0`; or wait 30 s after start-up. |
| $CO_2$ too low or too high in clean air | Forced recalibration outdoors (SCD41 section). |
| Temperature higher than the room | The box warms itself; compare boxes side by side. |
| Logs are not next to the application | The application folder is not writable; look in `MagicBox\logs` in `Documents`. |
| Excel opens the CSV in a single column | Export as .xlsx (the default) or use the CSV export with `;`. |

Appendix A

# Quick reference — JSON formats

## A.1 Multi-sensor packet (1 Hz)

```
{
  "id":    "BOX_A1B2",
  "ts_ms": 123456,
  "data": {
    "lux":     312.5,
    "white":   1840,
    "raw_als": 1821,
    "T":       22.34,
    "H":       58.12,
    "P":       1013.28,
    "ALT":     0.3
  }
}
```

## A.2 IMU packet (50 Hz)

```
{
  "id":     "BOX_A1B2",
  "ts_ms":  123476,
  "imu_hz": 50,
  "imu":    [[-0.023, 0.014, 0.998, 0.152, -0.089, 0.041]]
}
```

## A.3 SCD41 + BME280 packet

```
{
  "id":    "BOX_C3D4",
  "ts_ms": 5000,
  "data": {
    "co2":      842,
    "TEMP_SCD": 23.1,
    "HUM_SCD":  55.4,
    "T":        22.98,
    "H":        57.2,
    "P":        1011.45
  }
}
```

## Appendix B

# I²C wiring

All sensors share the same I²C bus, through the box's three sockets (at most three sensors at a time, section 2.1). A common GND is mandatory.

| Sensor | SDA | SCL | VCC |
|---|---|---|---|
| MPU-6500 | GPIO 21 | GPIO 22 | 3.3 V |
| BME280 | GPIO 21 | GPIO 22 | 3.3 V |
| VEML7700 | GPIO 21 | GPIO 22 | 3.3 V |
| SCD41 | GPIO 21 | GPIO 22 | 3.3 V |

| Sensor | I²C address (7-bit) |
|---|---|
| MPU-6500 | `0x68` (AD0=GND) or `0x69` (AD0=VCC) |
| BME280 | `0x76` or `0x77` (SDO pin) |
| VEML7700 | `0x10` (fixed) |
| SCD41 | `0x62` (fixed, Sensirion) |

**Critical constraint**: the SCD41 limits the bus to a maximum of 100 kHz. At 400 kHz, the MPU-6500 "disappears" from the bus due to the SCD41's clock-stretch interference.